Related Work and Research Gap: A Graph Neural Network Approach for EMG-Based Gesture Recognition on NinaPro DB1

Group 04

Related Work

Table 1: Summary of related work (2022–2026) on GNN/GCN-based physiological-signal gesture recognition.

Authors (Year)	Dataset	Application Area	GNN Type	Strengths	Limitations / Gap
Xu, Chen, Ruan, & Zhang [1]	HD-sEMG (multi-subject)	Cross-user EMG pattern / gesture recognition	Spatio-temporal GCN (CNN-MSTGCN)	Combines CNN feature extraction with a multi-scale spatio-temporal graph module; targets cross-user generalization	Channel-graph topology is fixed rather than learned per gesture
Lee et al. [2]	Custom stretchable 18-gesture array	Static/dynamic hand gesture recognition	GNN on flexible-array electrodes	∼97% accuracy pairing a stretchable sensor array with a spatially-aware graph model	Custom hardware/dataset limits comparability to public EMG benchmarks
Massa, Riboni, & Nazarpour [3]	HD-sEMG, 65 gestures, 20 subjects	Fine-grained gesture-intention recognition	GNN + GNNExplainer topology refinement (xAI-GNN)	Uses explainable AI to prune uninformative edges, improving interpretability and efficiency	Preliminary version reported ∼8.75% average error with high variance across gestures
Lee, Jiang, Yang, Yang, & Zhao [4]	Five public EMG datasets	Generalizable, interpretable gesture classification	Spatiotemporal GCN	Graph encodes both spatial electrode proximity and temporal adjacency; state-of-the-art across datasets	Prioritizes cross-dataset generalization over single-subject, low-channel deployment
Massa, Riboni, & Nazarpour [5]	HD-sEMG, 65 gestures, 20 subjects	Movement-intention recognition (initial study)	Baseline GNN on HD-EMG channel graph	Early proof-of-concept that GNNs are viable for raw-EMG-channel gesture decoding	Simple, non-adaptive graph topology; only two HD-EMG arrays tested
Vipulanandan, Premaratne, & Murthi [6]	Myoband sEMG, 8 electrodes, 8 subjects	Real-time hand gesture recognition	GNN on correlation-based signal graph	99% accuracy; graph built directly from inter-channel signal correlation, not fixed layout	Evaluated on only 5 gesture classes and a small subject pool

Table 1 – continued

Authors (Year)	Dataset	Application Area	GNN Type	Strengths	Limitations / Gap
Li, Ji, Yu, Li, Jin, Liu, Bai, & Ye [7]	EEG + EMG, stroke rehabilitation cohort	Multimodal stroke rehabilitation BCI	GIN (Graph Isomorphism Network) on heterogeneous graph	Combines EEG and EMG as a heterogeneous graph using mutual information for edges; strong gains over baselines	Heterogeneous EEG-EMG graph adds complexity not needed for EMG-only setups
Quadrelli, Canepa, Di Domenico, Boccardo, Chiappalone, & Laffranchi [8]	HD-EMG interfaces (review scope)	Upper-limb myoelectric prosthetic control	Review of GNN and spatial-ML approaches	Surveys how spatial/graph-based ML exploits HD-EMG electrode layout to improve intention detection	Mini-review scope; does not benchmark a single model on a standard low-channel dataset
Klepl, He, Wu, Blackburn, & Sarrigiannis [9]	Cross-dataset (EEG classification survey)	Systematic review of GNN-based physiological signal classification	Survey of spectral and spatial GNN variants	Identifies that spectral graph convolutions dominate current GNN-based biosignal classifiers	Focused on EEG; does not cover low-channel sEMG graph construction specifically
Hou, Jia, Lun, Hao, Shi, Li, Zeng, & Lv [10]	EEG motor-imagery benchmark datasets	Time-resolved motor-imagery decoding	GCNs-Net (graph convolution + pooling)	Learns generalized features respecting electrode topology; robust across individual variability	Designed for EEG electrode topology, not directly validated on sparse EMG channel graphs

Research Gap

- Existing GNN/GCN work on EMG gesture recognition is concentrated on **high-density (HD) sEMG** setups with dozens to over a hundred electrodes [1, 3, 4]; standard **low-channel (10-electrode)** surface EMG benchmarks such as NinaPro DB1 remain largely unexamined with GNNs.
- Graph topology in most prior work is fixed by **physical electrode adjacency** rather than learned from **inter-channel signal correlation**, which is the more relevant cue when only 10 sparsely-placed channels are available [5].
- Custom-hardware studies [2] report strong accuracy but on non-public, non-comparable datasets, limiting reproducibility against standard EMG benchmarks.
- Most GNN-based bio-signal survey and review work to date centers on **EEG** [9, 10], with comparatively few systematic studies applying the same graph-construction principles to low-channel sEMG.
- None of the reviewed studies explicitly address **severe rest-vs-gesture class imbalance** within the graph construction or training strategy itself, despite this being a dominant characteristic of raw EMG streams.
- **This work** closes this gap by building a correlation-driven channel graph from a standard low-density NinaPro configuration, evaluated under a subject-independent, leakage-free split with explicit class-imbalance handling.

References

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